Joshua Aney

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Education

Montana State University — Bozeman, MT
Bachelor of Science, Computer Science, GPA: 3.96 — *May 2026*

Skills & Coursework

Languages: Python, MATLAB, Java, C, SQL, C#
Frameworks: PyTorch, TensorFlow, WebDataset, .NET Core, SLURM, NumPy, Matplotlib
Tools: Linux, Jetbrains Products, GitHub, VSCode, Adobe Illustrator, Overleaf, Claude Code

Experience

Research Technologist — June 2026 - Current, (Intern Summers 2024, 2025)
CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic — *Rochester, MN*

- **Tools**: MATLAB, Signal Processing, Adobe Illustrator, Linear Algebra, Technical Writing
- Performed computational motor and language mapping on epilepsy patients to localize task-active brain regions.
- Recorded sEEG experiments in the Epilepsy Monitoring Unit and preprocessed raw BCI2000 data for analysis.
- Developed a novel method for dynamic brain network identification from evoked responses.
- Modified Canonical Response Parameterization for better temporal feature extraction from auditory evoked responses.

Undergraduate Research Assistant — January 2026 - June 2026
HACR Lab (Harnessing Automation in Cybersecurity Reasoning), Montana State University — *Bozeman, MT*

- **Tools**: Pytorch, Ghidra, High Performance Computing, SLURM
- Built pipelines to produce control flow graphs from compiled binaries for code embedding generation.
- Designed PyTorch Graph Neural Network architectures to leverage the features of control flow graphs.
- Ran machine learning workloads on Tempest, MSU's high performance computing system.

Teaching and Curriculum Development — Fall 2025, Spring 2026
Montana State University — *Bozeman, MT*

- **Tools**: Java, HTMX, SQLite, HTML, CSS
- Co-designed a semester-long web development course project template with a professor and three students.
- Implemented core functionality (MVC controllers, auth, multi-step workflows) for a veterinary clinic system.
- Structured a progressive assignment sequence so students could add controllers from established codebase patterns.
- Assisted students in learning core data structures algorithms (trees, hashmaps, depth first search, etc.)
- Taught students how to debug in their IDE's.

Technical Projects

Machine Learning and AI Models from Scratch | *Python, NumPy, Linear Algebra, Latex* — 2024-2026

- Implemented from scratch: Supervised, unsupervised, and rienforcement learning algorithms (Naïve Bayes, KNN, K-Means, DBSCAN, Multi-layer Perceptron, Q-Learning, SARSA, Value Iteration) and optimizers (backpropagation, genetic algorithm, differential evolution, particle swarm optimization, simulated annealing).
- Implemented a First-Order-Logic engine to perform inference on knowledge bases using resolution.

Bozeman Bike Trails Application | *C#, Razor Pages, SQL Server, Mapbox* — 2024

- Developed a web application to track user-ridden bike trail GPS data

Hand Tracking / ASL Alphabet Detection | *Python, TensorFlow, MediaPipe* — 2024

- Created a pipeline to collect data, train a dense neural network, and provide real time predictions for video inputs.

Extracurricular Activities & Leadership

President — August 2023 – Present
Montana State University Tennis Club — *Bozeman, MT*

Section Leader — April 2025 – January 2026
Montana State University Drumline — *Bozeman, MT*